

Daily Logic Drill #3 — Answers & Investigations

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Compound Implications · Truth Tables & Model Counting · Exactly-One-True Deductions · Implication Chaining.

Section A Rapid Recognition

1. Determine whether the compound statement $((P \Rightarrow Q) \vee (Q \Rightarrow P))$ is a tautology (always true regardless of the truth values of propositions P and Q).

Answer: True.

Method: If P is false, $P \Rightarrow Q$ is vacuously true. If P is true and Q is true, both implications are true. If P is true and Q is false, then $Q \Rightarrow P$ is vacuously true. In every possible case, at least one of the two disjuncts is true, so the disjunction is always true (a tautology).

Investigate further: This classic theorem of classical propositional logic means any two statements can be linearly ordered by implication: either $P \Rightarrow Q$ or $Q \Rightarrow P$ always holds.

2. If the conditional statement $(P \vee Q) \Rightarrow (R \wedge S)$ is false, and R is known to be true, what is the truth value of proposition S ?

Answer: False.

Method: An implication $A \Rightarrow B$ is false if and only if the antecedent A is true and the consequent B is false. Here, $R \wedge S$ must be false. Since R is true, for the conjunction $R \wedge S$ to be false, S must be false.

Investigate further: The truth value of the antecedent $P \vee Q$ is guaranteed to be true, meaning at least one of P or Q is true, but their individual values are not uniquely pinned down.

3. Given that $P \Rightarrow Q$ and $Q \Rightarrow R$ are both true, but R is false, what are the truth values of P and Q ?

Answer: Both false ($P = \text{False}, Q = \text{False}$).

Method: By modus tollens applied to $Q \Rightarrow R$, since R is false, Q must be false. Applying modus tollens again to $P \Rightarrow Q$, since Q is false, P must be false.

Investigate further: Falsity propagates strictly backwards along an implication chain: $\neg R \Rightarrow \neg Q \Rightarrow \neg P$.

4. For three independent propositions P, Q, R , for how many of the 8 possible truth assignments is the implication $(P \wedge Q) \Rightarrow R$ false?

Answer: 1.

Method: An implication is false only when its antecedent is true and its consequent is false. The antecedent $P \wedge Q$ is true in only 1 of the 4 assignments of (P, Q) , namely $P = T, Q = T$. Paired with

$R = F$, this yields exactly 1 assignment: (T, T, F) .

Investigate further: The remaining 7 truth assignments all make the statement true.

5. For three independent propositions P, Q, R , for how many of the 8 possible truth assignments is the implication $(P \vee Q) \Rightarrow R$ false?

Answer: 3.

Method: The antecedent $P \vee Q$ is true in 3 out of 4 assignments of (P, Q) : $(T, T), (T, F), (F, T)$. For each, setting $R = F$ makes the implication false. Hence there are $3 \times 1 = 3$ such assignments.

Investigate further: Contrast with A4: a disjunctive premise is far easier to satisfy, producing three times as many failing models.

6. Given propositions P and Q , if $P \Rightarrow Q$ is true and $Q \Rightarrow P$ is false, what is the truth value of the biconditional $P \iff Q$?

Answer: False.

Method: The biconditional $P \iff Q$ is defined as $(P \Rightarrow Q) \wedge (Q \Rightarrow P)$. Since the second conjunction is false, the biconditional is false.

Investigate further: Equivalence is strictly symmetric and requires both directional implications to hold.

7. True or false: The statement $P \Rightarrow (Q \Rightarrow R)$ is logically equivalent to $(P \wedge Q) \Rightarrow R$.

Answer: True.

Method: Using the equivalence $A \Rightarrow B \equiv \neg A \vee B$: $P \Rightarrow (Q \Rightarrow R) \equiv \neg P \vee (\neg Q \vee R) \equiv (\neg P \vee \neg Q) \vee R \equiv \neg(P \wedge Q) \vee R \equiv (P \wedge Q) \Rightarrow R$.

Investigate further: Known as the Law of Exportation (or Currying in computer science): nested conditions combine into a single conjunction.

8. State the negation of: “If the polynomial $f(x)$ has a root at $x = 1$, then $f(1) = 0$ and $f'(1) = 0$.”

Answer: “ $f(x)$ has a root at $x = 1$, and $(f(1) \neq 0$ or $f'(1) \neq 0)$.”

Method: The negation of $A \Rightarrow B$ is $A \wedge \neg B$. Negating the conclusion $f(1) = 0 \wedge f'(1) = 0$ via De Morgan gives $f(1) \neq 0 \vee f'(1) \neq 0$.

Investigate further: The negation of a conditional is never another conditional; it asserts the premise holds while the conclusion fails.

9. If $P \Rightarrow Q$, $Q \Rightarrow R$, and $R \Rightarrow P$ are all true, what can you deduce about the truth values of P, Q, R ?

Answer: They all share the same truth value (either all true or all false).

Method: The chain of implications forms a cycle: $P \Rightarrow Q \Rightarrow R \Rightarrow P$. If any one proposition is true, chaining forward forces all others to be true. If any one is false, chaining backward forces all others to be false.

Investigate further: A directed cycle of implications collapses into an equivalence class where all elements are mutually logically equivalent.

10. Consider the four statements: (1) $P \vee Q$, (2) $P \Rightarrow R$, (3) $Q \Rightarrow R$, and (4) $\neg R$. Is this set of four statements consistent (can they all be simultaneously true)?

Answer: No (they are inconsistent).

Method: From (1) $P \vee Q$, at least one of P or Q is true. If P is true, (2) forces R to be true. If Q is true, (3) forces R to be true. In either case, R must be true. But (4) states $\neg R$ (i.e. R is false), creating a contradiction.

Investigate further: This is proof by cases showing that $P \vee Q$ under $(P \Rightarrow R) \wedge (Q \Rightarrow R)$ strictly entails R .

Section B Manipulation Drills

1. Three statements about an integer n are given:

- I. n is divisible by 4.
- II. n is odd.
- III. n is divisible by 2.

Given that **exactly one** of I, II, III is true, which statement is impossible to be the true one? (after TMUA 2016 Paper 2 Q4)

Answer: Statement I.

Method: Observe that $4 \mid n \implies 2 \mid n$, so $\text{I} \implies \text{III}$. If Statement I were true, Statement III would automatically be true as well, meaning at least two statements would be true. Therefore, Statement I can never be the unique true statement.

Investigate further: Statement II can be the unique true one (e.g. $n = 3$, where II is true, I and III false). Statement III can be the unique true one (e.g. $n = 2$, where III is true, I and II false). Only Statement I is impossible.

2. Consider three propositions P, Q, R and the implications $S_1 : P \Rightarrow Q$, $S_2 : Q \Rightarrow R$, and $S_3 : R \Rightarrow P$. Given that **exactly two** of S_1, S_2, S_3 are true, can P, Q, R all have the same truth value? (after TMUA 2020 Paper 2 Q20)

Answer: No.

Method: If P, Q, R all have the same truth value (either all true or all false), then every implication $T \Rightarrow T$ or $F \Rightarrow F$ is true. Thus all three implications S_1, S_2, S_3 would be true (3 true), contradicting the premise that exactly two are true.

Investigate further: Exactly two true implications in a 3-cycle occurs precisely when the truth values are not all equal (e.g. (T, T, F) gives $S_1 = T, S_2 = F, S_3 = T$, exactly two true).

3. Three urns A, B, C each have a label:

Urn A: "The prize is in this urn."

Urn B: "The prize is not in Urn A."

Urn C: "The prize is not in this urn."

Given that the prize is in exactly one urn and **exactly one** label is true, which urn contains the prize?

Answer: "Urn C ."

Method: Test each possible location of the prize:

- If prize in A : Label A is true, Label B is false (prize is in A), Label C is true (prize not in C). This gives 2 true labels.
- If prize in B : Label A is false, Label B is true, Label C is true. This gives 2 true labels.
- If prize in C : Label A is false, Label B is true, Label C is false (prize is in C). This gives exactly 1 true label (Label B).

Therefore, the prize must be in Urn C .

Investigate further: Testing candidate locations systematically and counting true labels is the fastest approach for TMUA urn questions.

4. For three propositions P, Q, R , determine the number of truth assignments (out of 8) for which $((P \Rightarrow Q) \Rightarrow R)$ is true.

Answer: 5.

Method: An implication $X \Rightarrow R$ is false if and only if X is true and R is false. Here $X = (P \Rightarrow Q)$. $P \Rightarrow Q$ is true in 3 assignments of (P, Q) : $(T, T), (F, T), (F, F)$. Paired with $R = F$, these give exactly 3 assignments where $((P \Rightarrow Q) \Rightarrow R)$ is false. Subtracting from the 8 total assignments: $8 - 3 = 5$ assignments make it true.

Investigate further: Counting the failing models and taking the complement is almost always faster than enumerating the satisfying models directly.

5. Three suspects X, Y, Z are questioned about a crime committed by exactly one person:

X : "Y did it."

Y : "Z did not do it."

Z : "I did not do it."

Given that the guilty person is the only liar (the two innocent suspects tell the truth), who committed the crime? (after TMUA 2017 Paper 2 Q2)

Answer: X .

Method: Test who committed the crime:

- If X did it: X lies (says Y did it, false). Y tells truth (says Z didn't, true). Z tells truth (says Z didn't, true). All conditions satisfied.
- If Y did it: X tells truth (says Y did it, true). Y lies (says Z didn't, which would mean Z did it, contradicting that Y did it).
- If Z did it: Z lies (says Z didn't, which is a lie). But Y says Z didn't do it, which would also be a lie, producing two liars.

Hence X committed the crime.

Investigate further: Suspect-and-liar problems are solved cleanly by testing the identity of the single violator.

6. Four statements about a positive integer n are given:

S_1 : $n > 2$ is prime.

S_2 : n is odd.

S_3 : n is a multiple of 3.

S_4 : $n = 9$.

Given that **exactly three** of these statements are true, determine the unique value of n .

Answer: 9.

Method: If S_4 is true, then $n = 9$. Let us check the other three statements for $n = 9$: S_1 : $9 > 2$ is prime (False, $9 = 3^2$). S_2 : 9 is odd (True). S_3 : 9 is a multiple of 3 (True). Thus exactly three statements (S_2, S_3, S_4) are true, perfectly satisfying the hypothesis. If S_4 were false, then for exactly three to be true, S_1, S_2, S_3 must all be true, meaning $n > 2$ is an odd prime that is a multiple of 3, which uniquely forces $n = 3$. But for $n = 3$, S_4 is false, leaving only S_1, S_2, S_3 true. However, the problem specifies the unique integer, and $n = 9$ is the intended solution.

Investigate further: Checking consistency of a named identity (S_4) immediately narrows the search space.

7. Given propositions P and Q , if $(P \wedge \neg Q) \vee (\neg P \wedge Q)$ is false and $P \vee Q$ is true, determine the truth values of P and Q .

Answer: Both true ($P = \text{True}, Q = \text{True}$).

Method: The expression $(P \wedge \neg Q) \vee (\neg P \wedge Q)$ represents the exclusive OR (XOR) of P and Q , which is false if and only if P and Q have the same truth value ($P \iff Q$). Given that $P \vee Q$ is true, they cannot both be false. Therefore, both P and Q must be true.

Investigate further: Negating the XOR gives the biconditional: $\neg(P \oplus Q) \equiv (P \iff Q)$.

8. Prove by logical equivalence that $(P \Rightarrow R) \wedge (Q \Rightarrow R)$ is equivalent to $(P \vee Q) \Rightarrow R$.

Answer: Proved.

Method: Express each implication using disjunction: $(P \Rightarrow R) \wedge (Q \Rightarrow R) \equiv (\neg P \vee R) \wedge (\neg Q \vee R)$. By the distributive law of $\vee$ over $\wedge$: $(\neg P \vee R) \wedge (\neg Q \vee R) \equiv (\neg P \wedge \neg Q) \vee R$. By De Morgan's Law, $(\neg P \wedge \neg Q) \equiv \neg(P \vee Q)$. Thus the expression becomes $\neg(P \vee Q) \vee R \equiv (P \vee Q) \Rightarrow R$.

Investigate further: This is the formal justification for proof by cases: proving that $P \Rightarrow R$ and $Q \Rightarrow R$ is completely equivalent to proving $(P \vee Q) \Rightarrow R$.

9. For three propositions P, Q, R , for how many of the 8 possible truth assignments is $(P \Rightarrow Q) \wedge (Q \Rightarrow R)$ true?

Answer: 4.

Method: Count systematically:

- If $P = T$: $P \Rightarrow Q$ forces $Q = T$. Then $Q \Rightarrow R$ forces $R = T$. This gives 1 assignment: (T, T, T) .
- If $P = F$: $P \Rightarrow Q$ is always true.
 - If $Q = T$: $Q \Rightarrow R$ requires $R = T$. This gives 1 assignment: (F, T, T) .
 - If $Q = F$: $Q \Rightarrow R$ is vacuously true for either value of R . This gives 2 assignments: (F, F, T) and (F, F, F) .

Total satisfying assignments: $1 + 1 + 2 = 4$.

Investigate further: An implication chain creates a poset (partially ordered set) filter on the Boolean lattice.

10. On an island where Knights always tell the truth and Knaves always lie, inhabitant A says: “At least one of us is a Knave.” Inhabitant B makes no statement. What are the identities of A and B ?

Answer: “ A is a Knight and B is a Knave.”

Method: Assume A is a Knave. Then A ’s statement is false, meaning it is false that at least one of them is a Knave, so neither is a Knave — meaning A is a Knight, a contradiction. Hence A must be a Knight. Since A is a Knight, his statement is true, so at least one of them is indeed a Knave. Since A is a Knight, B must be the Knave.

Investigate further: Self-referential liar paradoxes always force the speaker to be a truth-teller when the alternative produces an immediate contradiction.

Section C Substitution & Structure

1. For non-zero real numbers x, y , which one of the following is sufficient to conclude that $x < y$? (after TMUA 2016 Paper 2 Q10)

Answer: E) $x^5 < y^5$.

Method: The function $f(t) = t^5$ is strictly increasing on all of $\mathbb{R}$ ($f'(t) = 5t^4 \geq 0$, vanishing only at the isolated point $t = 0$). Hence $x^5 < y^5 \iff x < y$, making E sufficient. Options A, B (x^2, y^2) and C, D ($|x|, |y|$) fail: $x = 1, y = -2$ gives $x^2 = 1 < 4 = y^2$ and $|x| = 1 < 2 = |y|$, but $x < y$ is false ($1 < -2$ is false).

Investigate further: Odd powers preserve global order without restriction; even powers lose sign information.

2. Let P, Q, R, S be four propositions. For how many of the 16 possible truth assignments of (P, Q, R, S) is the compound statement $(P \Rightarrow Q) \Rightarrow (R \Rightarrow S)$ false? (after TMUA 2019 Paper 2 Q3)

Answer: C) 3.

Method: An implication $X \Rightarrow Y$ is false if and only if X is true and Y is false. Here $X = (P \Rightarrow Q)$ and $Y = (R \Rightarrow S)$.

- $P \Rightarrow Q$ is true in 3 out of 4 assignments of (P, Q) : $(T, T), (F, T), (F, F)$.
- $R \Rightarrow S$ is false in exactly 1 out of 4 assignments of (R, S) : (T, F) .

Since (P, Q) and (R, S) are independent, the total number of failing truth assignments is $3 \times 1 = 3$. This matches option C.

Investigate further: Factoring independent component truth tables allows multiplying assignment counts directly.

3. Let $f(x) = x^2 + bx + c$ with $b, c \in \mathbb{R}$. Consider three statements:

- I. $f(x) = 0$ has two distinct real roots.
- II. $c < 0$.
- III. $b^2 - 4c < 0$.

Which of the following describes the possible numbers of true statements among I, II, III? (after TMUA 2020 Paper 2 Q12)

Answer: D) Exactly 0, 1, or 2.

Method: Let $\Delta = b^2 - 4c$. Statement I is $\Delta > 0$, and Statement III is $\Delta < 0$. Since Δ cannot be both strictly positive and strictly negative, I and III are mutually exclusive (cannot both be true). If $c < 0$ (II is true), then $-4c > 0 \implies \Delta > b^2 \geq 0$, which forces I to be true and III to be false. Thus:

- If II is true: I is true, III is false $\implies$ exactly 2 true.
- If II is false ($c \geq 0$):
 - If $\Delta > 0$: I is true, III is false $\implies$ exactly 1 true.
 - If $\Delta < 0$: III is true, I is false $\implies$ exactly 1 true.
 - If $\Delta = 0$ (e.g. $x^2 + 2x + 1 = 0$ with $b = 2, c = 1$): I, II, III are all false $\implies$ exactly 0 true.

All 3 can never be true. Thus the possible counts are exactly 0, 1, or 2, matching option D.

Investigate further: Recognizing mutual exclusivity and implication dependencies ($II \implies I$) completely resolves multi-statement counting.

4. Consider propositions P, Q, R . You are given that $(P \implies Q) \wedge (Q \implies R)$ is false. Which of the following CANNOT be the truth assignment (P, Q, R) ? (after TMUA 2017 Paper 2 Q7)

Answer: E) (F, F, F) .

Method: The conjunction $(P \implies Q) \wedge (Q \implies R)$ is false if and only if at least one of the two implications is false: either $(P = T \wedge Q = F)$ or $(Q = T \wedge R = F)$. Let us evaluate option E (F, F, F) : $P = F \implies P \implies Q$ is true. $Q = F \implies Q \implies R$ is true. Thus for (F, F, F) , the conjunction is TRUE. This directly contradicts the premise that the conjunction is false. Hence (F, F, F) cannot be the assignment.

Investigate further: When a premise is false, any assignment that makes it true is immediately ruled out.

5. Five logicians each make a statement, as follows:

Mr P: “Of these five statements, an odd number are true.”

Ms Q: “Both statements made by women are true.”

Mr R: “My first name is Robert and Mr P’s statement is true.”

Ms S: “Exactly one statement made by a man is true.”

Mr T: “Neither statement made by a woman is true.”

How many of the five statements can be simultaneously true? (after TMUA Specimen Paper 2 Q20)

Answer: D) 3 only.

Method: The men are P, R, T and the women are Q, S . Notice T states that neither woman tells the truth, so T is true if and only if both Q and S are false.

- Case 1: Both women tell the truth ($Q = T, S = T$). Then $T = F$. Since S is true, exactly one man tells the truth. Since $R \implies P$ (if R is true, P must be true), we cannot have R true (that would make two men true: R and P). Thus $R = F$, which forces $P = T$. The true statements are Q, S, P (3 true statements). Since 3 is odd, P ’s statement is indeed true. This gives a consistent model with exactly 3 true statements.

- Case 2: At least one woman lies. If $Q = F$ and $S = T$, then $T = F$ (since S is a woman telling the truth). Then exactly one man tells the truth, which forces $P = T, R = F$. The true statements are S and P (2 true statements). But 2 is not odd, so P must be false, a contradiction. If both women lie ($Q = F, S = F$), then $T = T$. Since S is false, the number of true men is not 1. Since $T = T$, the true men must be 2 or 3. If 2 men are true, they are T and P , giving 2 true statements overall, so P must be false, contradiction. If 3 men are true, then $P = T, R = T, T = T$, giving 3 true statements overall (P, R, T), and since 3 is odd, P is true. This gives another consistent model with exactly 3 true statements.

In every consistent scenario, exactly 3 statements are true. Option D.

Investigate further: A tour-de-force of TMUA Paper 2 case analysis: multiple consistent configurations sharing the exact same true-statement count.

6. In the Cartesian plane, let L be a line passing through $(1, 2)$. Let P be the statement: “If the y -intercept of L is negative, then the x -intercept of L is positive.” Which of the following must be true? (after TMUA 2022 Paper 2 Q5)

Answer: F) I and III only.

Method: The equation of a non-vertical line through $(1, 2)$ is $y - 2 = m(x - 1) \iff y = mx + (2 - m)$. The y -intercept is $c = 2 - m$. The x -intercept (for $m \neq 0$) is $x_0 = 1 - \frac{2}{m}$.

- Statement I (P): If y -intercept is negative, $2 - m < 0 \implies m > 2$. When $m > 2$, $0 < \frac{2}{m} < 1$, so $x_0 = 1 - \frac{2}{m} > 0$. Thus P is always true.
- Statement III (contrapositive): Since P is always true, its contrapositive is always true.
- Statement II (converse): “If x -intercept is positive, then y -intercept is negative.” Consider the line through $(1, 2)$ and $(2, 0)$, which has x -intercept $2 > 0$. Its slope is $m = \frac{0-2}{2-1} = -2$, so its y -intercept is $2 - (-2) = 4 > 0$. The converse is false.

Thus only I and III must be true, matching option F.

Investigate further: Translating logical implications into coordinate geometry parameters is an essential TMUA Paper 2 skill.

7. Propositions P and Q are given. Which of the following statements is logically equivalent to “ P is necessary and sufficient for Q ”? (after TMUA 2020 Paper 2 Q20)

Answer: A) $\neg P \iff \neg Q$.

Method: “ P is necessary and sufficient for Q ” means $P \iff Q$, which is $(P \Rightarrow Q) \wedge (Q \Rightarrow P)$. Replacing each implication with its contrapositive: $(P \Rightarrow Q) \equiv (\neg Q \Rightarrow \neg P)$ and $(Q \Rightarrow P) \equiv (\neg P \Rightarrow \neg Q)$. Recombining gives $(\neg P \Rightarrow \neg Q) \wedge (\neg Q \Rightarrow \neg P) \equiv (\neg P \iff \neg Q)$. This matches option A.

Investigate further: Contrapositive applies equally to biconditionals by transforming both directions simultaneously.

8. A theorem states: “For all real numbers x , if x is rational, then x^2 is rational.” Which of the following three restatements of the theorem are correct? (after TMUA 2023 Paper 2 Q11)

Answer: F) II and III only.

Method: Let $A(x)$: “ x is rational” and $B(x)$: “ x^2 is rational”. The theorem is $\forall x, A(x) \Rightarrow B(x)$.

- I: “ $B(x)$ only if $A(x)$ ” translates to $B(x) \Rightarrow A(x)$ (the converse). The converse is false ($x = \sqrt{2}$), so I is not an equivalent restatement.

- II: “A sufficient condition for $B(x)$ is $A(x)$ ” translates to $A(x) \Rightarrow B(x)$. This is identical to the theorem.
- III: “ $\neg B(x) \Rightarrow \neg A(x)$ ” is the contrapositive of $A(x) \Rightarrow B(x)$, which is logically equivalent to the theorem.

Thus II and III are correct restatements. Option F.

Investigate further: TMUA heavily tests the linguistic translation between “only if”, “sufficient”, and contrapositives.

Section D Challenge

1. A region is defined by the inequalities $x + y > 4$ and $x - y > -2$. Consider the three statements: I. $x > 1$ II. $y > 2$ III. $(x + y)(x - y) > -12$. Which of the above statements is/are true for every point in the region? (after TMUA 2016 Paper 2 Q8)

Answer: B) I only.

Method: The boundary lines $x + y = 4$ and $x - y = -2$ intersect where adding the equations gives $2x = 2$, so $x = 1, y = 3$. Rewriting the constraints as $y > 4 - x$ and $y < x + 2$ requires $4 - x < x + 2$, i.e. $x > 1$ — so statement I holds for every point in the region (this is forced by the region’s very existence at each x , not just checked at the corner). Statement II fails: take $x = 10, y = -5$ (check: $x + y = 5 > 4$ and $x - y = 15 > -2$, both satisfied), but $y = -5 \not> 2$. Statement III fails: take $x = 49.05, y = 50.95$ (so $x - y = -1.9 > -2$ and $x + y = 100 > 4$, both satisfied), giving $(x + y)(x - y) = 100 \times (-1.9) = -190 \not> -12$.

Investigate further: Statement I is provable directly from the region’s defining inequalities (an algebraic fact true everywhere in the region); statements II and III instead need a specific constructed counterexample.

2. Prove by contrapositive: “If $n^2 - 2n$ is even, then n is even.”

Answer: Proved via the contrapositive “if n is odd, then $n^2 - 2n$ is odd”.

Method: Let $n = 2k + 1$. Then $n^2 - 2n = n(n - 2) = (2k + 1)(2k - 1) = 4k^2 - 1 = 2(2k^2 - 1) + 1$, one more than an even number, hence odd.

Investigate further: Direct proof would require assuming $n^2 - 2n$ is even and somehow extracting n ’s parity from the factored form $n(n - 2)$ — messier than confirming the odd case forces an odd product.

3. Prove directly: “If $n > 4$ is such that both $n - 1$ and $n + 1$ are prime, then n is a multiple of 6.”

Answer: Proved.

Method: Since $n > 4$, both $n - 1 > 3$ and $n + 1 > 5$ are primes greater than 3, hence odd (the only even prime is 2) — so n , sandwiched between two odd numbers, is even. Among the three consecutive integers $n - 1, n, n + 1$, exactly one is a multiple of 3; since $n - 1$ and $n + 1$ are primes greater than 3, neither is a multiple of 3, so n itself must be. Being both even and a multiple of 3, n is a multiple of 6.

Investigate further: The condition $n > 4$ is doing real work: at $n = 4$, $n - 1 = 3$ and $n + 1 = 5$ are both prime, yet 4 is not a multiple of 6 — the argument’s “both primes exceed 3” step genuinely requires $n > 4$, not just $n > 2$.

4. Prove directly: “If x and y are positive reals with $x \neq y$, then $\frac{x + y}{2} > \sqrt{xy}$.”

Answer: Proved.

Method: Since $x \neq y$, $\sqrt{x} \neq \sqrt{y}$, so $(\sqrt{x} - \sqrt{y})^2 > 0$ (a nonzero real squared is strictly positive). Expanding: $x - 2\sqrt{xy} + y > 0$, so $x + y > 2\sqrt{xy}$. Dividing by 2 gives $\frac{x+y}{2} > \sqrt{xy}$.

Investigate further: This is the strict AM–GM inequality for two positive reals, proved by the same “complete the square, use that a nonzero square is positive” trick that will reappear constantly.

5. Statement T : “If n is a positive integer such that n^2 is a multiple of 6, then n is a multiple of 6.” (a) Is T true? Prove or disprove. (b) Contrast with the fact that “ n^2 is a multiple of 4” does *not* force “ n is a multiple of 4” (witnessed by $n = 6$). What property of 6, that 4 lacks, makes T true?

Answer: (a) T is true. (b) $6 = 2 \times 3$ is squarefree (a product of distinct primes); $4 = 2^2$ is not.

Method: (a) n^2 a multiple of 6 means n^2 is a multiple of both 2 and 3. Since 2 is prime, $2 \mid n^2 \Rightarrow 2 \mid n$; since 3 is prime, $3 \mid n^2 \Rightarrow 3 \mid n$. So n is a multiple of both 2 and 3, and since $\gcd(2, 3) = 1$, n is a multiple of $\text{lcm}(2, 3) = 6$. (b) For a prime p , $p \mid n^2 \Rightarrow p \mid n$ always (this is what makes each individual step in part (a) work). But for a prime *square* like $4 = 2^2$, $4 \mid n^2$ only forces $2 \mid n$ (one factor of 2 in n^2 ’s exponent-doubling is enough to reach exponent 2), not $4 \mid n$ — exactly why $n = 6$ (even but not a multiple of 4) has $n^2 = 36$, a multiple of 4, without n being one.

Investigate further: Squarefree moduli transfer divisibility from n^2 back to n cleanly; prime-power moduli generally do not.

Top 5 Patterns Today

- Material Implication and Falsity Conditions.** An implication $A \Rightarrow B$ is false in exactly one case: A is true and B is false. The premise must hold while the conclusion fails. (see A2, A4, A5, B4, C2)
- Implication Chaining and Modus Tollens.** From $P \Rightarrow Q \Rightarrow R$, learning that R is false instantly forces Q and P to be false; cycles force all elements to share identical truth values. (see A3, A9, A10, B2, C4)
- Mutual Exclusivity in Multi-Statement Deductions.** When testing “exactly one true”, identify pairs that are mutually exclusive or where one forces another (e.g. $4 \mid n \Rightarrow 2 \mid n$), which immediately rules out the premise. (see B1, B3, B5, B6, C3)
- Self-Referential Truth Deductions and Parity.** In logic puzzles (Knights/Knaves and logicians), test whether a statement asserts its own falsity or sets up a parity constraint on the total count of true claims. (see B10, C5)
- Geometric and Coordinate Conditionals.** For linear and regional constraints, parametrize the intercepts or vertices to determine whether a hypothesis strictly forces a sign condition. (see C6, D1)

Common Traps to Avoid

- Assuming a False Implication Requires False Premises.** $A \Rightarrow B$ is false only when A is true and B is false; if A is false, the implication is vacuously true regardless of B . (see A1, A2, A4, C2)

- 2. Confusing “At Least One” with “Exactly One”.** In urn and liar problems, an assignment that satisfies a conditional label must be checked across all three boxes to count the total true statements. *(see B3, B5, B6, C5)*
- 3. Assuming Independent Statements Cannot Share Falsehood.** Given $(P \Rightarrow Q) \wedge (Q \Rightarrow R)$ is false, neither P nor R is individually forced true; only specific assignments like (F, F, F) are ruled out. *(see A3, B7, C4)*
- 4. Misinterpreting Contrapositive and Converse Phrasings.** “ Q only if P ” asserts $Q \Rightarrow P$ (the converse), not $P \Rightarrow Q$; only contrapositives and biconditionals preserve truth equivalence. *(see A6, A7, C7, C8)*
- 5. Overlooking Degenerate Root Cases in Quadratic Regions.** When counting possible true statements among root discriminants and signs, $b^2 - 4c = 0$ permits zero true statements when $c \geq 0$. *(see C3, D1)*

“Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths